First midterm FYS4480 – Quantum mechanics for many-particle systems, deadline October 26, at midnight. This midterm counts 25% of the final grade. Total score:100 points.

Department of Physics, University of Oslo, Norway

We present a simplified Hamiltonian consisting of an unperturbed Hamiltonian and a so-called pairing interaction term. It is a model which to a large extent mimics some central features of atomic nuclei, certain atoms and systems which exhibit superfluidity or superconductivity. To study this system we will use three of the methods of this course, in the order in which they have been developed in the lectures: full configuration interaction (FCI) theory, Hartree-Fock (HF) theory and the density matrix renormalization group (DMRG). The midterm is divided into three parts which follow this order and build on each other.

In Part I we use second quantization to set up the Hamiltonian matrix in the space of all Slater determinants of the model. Diagonalizing this matrix is the FCI method and provides us with the exact answer, which serves as the reference for everything that follows. We also look at what happens when the space of Slater determinants is truncated. In Part II we go to the opposite extreme and look for the best possible *single* Slater determinant, which is Hartree-Fock theory. In Part III we return to the exact FCI state and rewrite it as a matrix product state (MPS), and the Hamiltonian as a matrix product operator (MPO). The bond dimension χ of the MPS interpolates between a product state over the single-particle levels ($\chi = 1$) and the exact FCI state, and the DMRG is the variational method which finds the best MPS at a given χ. The three methods are thus different ways of parametrizing the same many-body wave function, and the aim of the midterm is to see how they are related for one and the same model. When setting up the Hamiltonian matrix you will need to solve an eigenvalue problem.

We define first the Hamiltonian, with a definition of the model space and the single-particle basis. Thereafter, we present the various exercises.

The Hamiltonian acting in the complete Hilbert space (usually infinite dimensional) consists of an unperturbed one-body part, $\hat{H}_0$, and a perturbation $\hat{V}$.

We limit ourselves to at most two-body interactions, our Hamiltonian is then represented by the following operators

$$\hat{H} = \sum_{\alpha\beta} \langle \alpha | h_0 | \beta \rangle a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | V | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma,$$

where $a_\alpha^\dagger$ and a_α etc. are standard fermion creation and annihilation operators, respectively, and $\alpha\beta\gamma\delta$ represent all possible single-particle quantum numbers. The full single-particle space is defined by the completeness relation $\hat{\mathbf{1}} = \sum_{\alpha=1}^{\infty} |\alpha\rangle\langle\alpha|$. In our calculations we will let the single-particle states $|\alpha\rangle$ be eigenfunctions of the one-particle operator $\hat{h}_0$. Note that the two-body part of the Hamiltonian contains anti-symmetrized matrix elements.

The above Hamiltonian acts in turn on various many-body Slater determinants constructed from the single-basis defined by the one-body operator $\hat{h}_0$. As an example, the two-particle model space $\mathcal{P}$ is defined by an operator

$$\hat{P} = \sum_{\alpha\beta=1}^{m} |\alpha\beta\rangle\langle\alpha\beta|,$$

where we assume that $m = \dim(\mathcal{P})$ and the full space is defined by

$$\hat{P} + \hat{Q} = \hat{\mathbf{1}},$$

with the projection operator

$$\hat{Q} = \sum_{\alpha\beta=m+1}^{\infty} |\alpha\beta\rangle\langle\alpha\beta|,$$

being the complement of $\hat{P}$.

Our specific model consists of N doubly-degenerate and equally spaced single-particle levels labelled by $p = 1, 2, \ldots$ and spin $\sigma = \pm 1$. These states are schematically portrayed in Fig. 1. The first two single-particle levels define a possible model space, indicated by the label $\mathcal{P}$. The remaining states span the excluded space $\mathcal{Q}$.

We write the Hamiltonian as

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where

$$\hat{H}_0 = \xi \sum_{p\sigma} (p-1) a^\dagger_{p\sigma} a_{p\sigma}$$

and

$$\hat{V} = -\frac{1}{2} g \sum_{pq} a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}.$$

Here, H_0 is the unperturbed Hamiltonian with a spacing between successive single-particle states given by ξ, which we will set to a constant value $\xi = 1$ without loss of generality (except in exercise 7 below, where we study the degenerate case $\xi = 0$). The two-body operator $\hat{V}$ has one term only. It represents the pairing contribution and carries a constant strength g. The pairing interaction is attractive for $g > 0$.

The indices $\sigma = \pm$ represent the two possible spin values. The interaction can only couple pairs and excites therefore only two particles at the time, as indicated by the rightmost four-particle state in Fig. 1. There one of the pairs is excited to the state with $p = 9$ and the other to the state $p = 7$. The two middle possibilities are not possible with the present model. We label single-particle states within the model space as hole-states. The single-particle states outside the model space are then particle states.

In our model we have kept both the interaction strength and the single-particle level as constants. In a realistic system like an atom or the atomic nucleus this is not the case.

Part I: Second quantization and full configuration interaction theory (40 points)

1. (15/100 points) Show that the unperturbed Hamiltonian $\hat{H}_0$ and $\hat{V}$ commute with both the spin projection $\hat{S}_z$ and the total spin $\hat{S}^2$, given by

$$\hat{S}_z := \frac{1}{2} \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}$$

and

$$\hat{S}^2 := \hat{S}_z^2 + \frac{1}{2}(\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+),$$

where

$$\hat{S}_\pm := \sum_p a^\dagger_{p\pm} a_{p\mp}.$$

This is an important feature of our system that allows us to block-diagonalize the full Hamiltonian. We will focus on total spin $S = 0$. In this case, it is convenient to define the so-called pair creation and pair annihilation operators

$$\hat{P}^+_p = a^\dagger_{p+} a^\dagger_{p-},$$

and

$$\hat{P}^-_p = a_{p-} a_{p+},$$

respectively.

Show that you can rewrite the Hamiltonian (with $\xi = 1$) as

$$\hat{H} = \sum_{p\sigma} (p-1) a^\dagger_{p\sigma} a_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{P}^+_p \hat{P}^-_q.$$

Show also that pair creation operators commute among themselves.

In this midterm we focus only on a system with no broken pairs. This means that the Hamiltonian can only link two-particle states in so-called spin-reversed states.

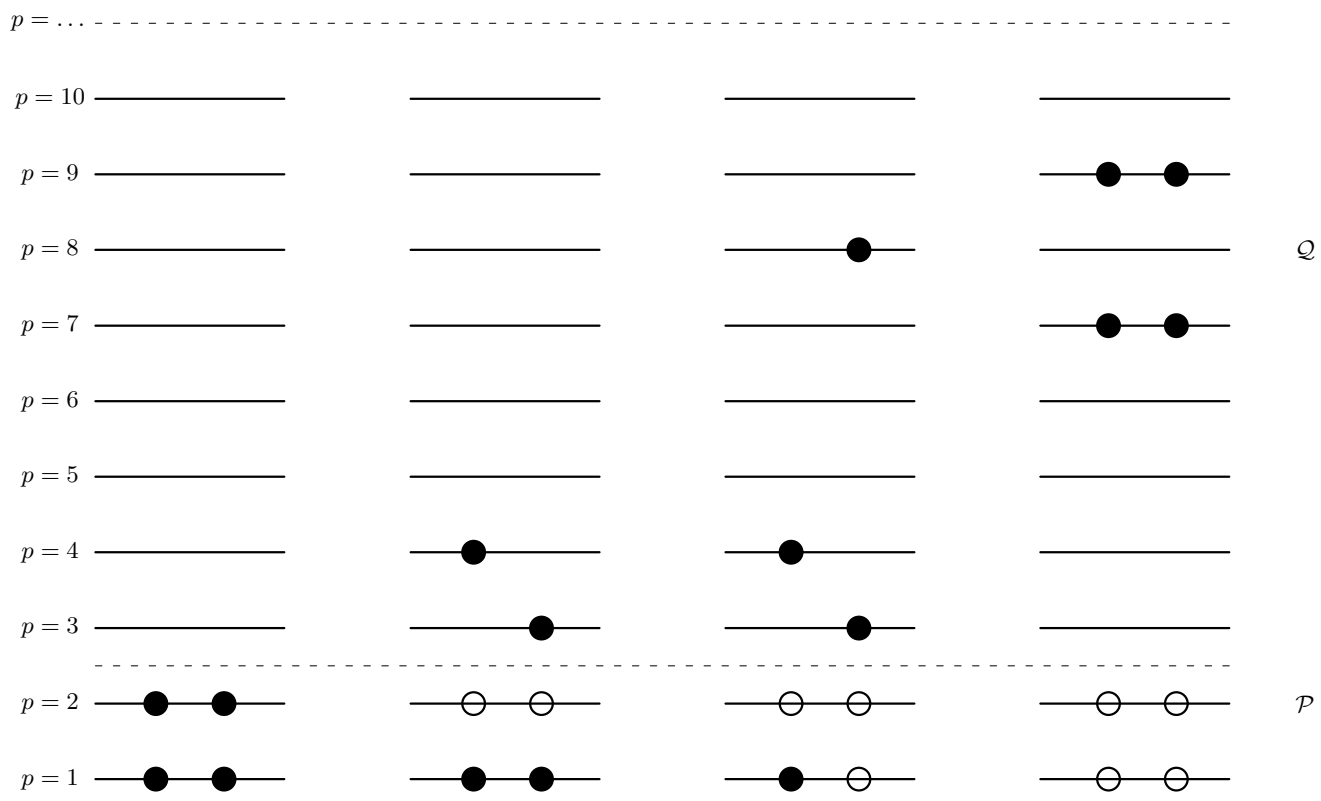


FIG. 1: Schematic plot of the possible single-particle levels with double degeneracy. The filled circles indicate occupied particle states while the empty circles represent vacant particle(hole) states. The spacing between each level p is constant in this picture. The first two single-particle levels define our possible model space, indicated by the label $\mathcal{P}$. The remaining states span the excluded space $\mathcal{Q}$. The first state to the left represents a possible ground state representation for a four-fermion system. In the second state to the left, one pair is broken. This possibility is however not included in our interaction.

2. (15/100 points) Construct thereafter the Hamiltonian matrix for a system with no broken pairs and total spin $S = 0$ for the case of the four lowest single-particle levels indicated in the Fig. 1. Our system consists of four particles only. Our single-particle space consists of only the four lowest levels $p = 1, 2, 3, 4$. You need to set up all possible Slater determinants. Find all eigenvalues by diagonalizing the Hamiltonian matrix. Vary your results for values of $g \in [-1, 1]$. We refer to this as the exact calculation. Comment the behavior of the ground state as function of g. Keep the ground-state eigenvector (not only the eigenvalue) for each value of g; you will need it in Part III.

3. (10/100 points)

   Instead of setting up all possible Slater determinants, construct only an approximation to the ground state (where we assume that the four particles are in the two lowest single-particle orbits only) which includes at most two-particle-two-hole excitations. Diagonalize this matrix and compare with the exact calculation and comment your results. Can you set up which diagrams this approximation corresponds to?

Part II: Hartree-Fock theory (15 points)

4. (15/100 points) We switch now to approximative methods, with Hartree-Fock theory first. Hereafter we will define our model space to consist of the single-particle levels $p = 1, 2$. The remaining levels $p = 3, 4$ define our excluded space. This means that our ground state Slater determinant consists of four particles which can be placed in the doubly degenerate orbits $p = 1$ and $p = 2$. Our first step is to perform a Hartree-Fock calculation with the pairing Hamiltonian. Write first the normal-ordered Hamiltonian with respect to the above reference state given by four spin 1/2 fermions in the single-particle levels $p = 1, 2$, and set up the standard Hartree-Fock equations for the above system (often called restricted Hartree-Fock due to the fact that we have an equal number of spin-orbitals).

   The standard way to perform a Hartree-Fock calculation is to expand the single-particle functions in a known basis and vary the coefficients, that is, the new function single-particle wave function $|p\rangle$ is written as a linear expansion in terms of a fixed basis ϕ

$$\psi_p = \sum_{\lambda} C_{p\lambda}\phi_{\lambda},$$

   This leads to a new Slater determinant which is related to the previous via a unitary transformation. We consider a Slater determinant built up of single-particle orbitals ϕ_λ where the indices λ refer to specific single-particle states.

   The unitary transformation

$$\psi_p = \sum_{\lambda} C_{p\lambda}\phi_{\lambda},$$

   brings us into the new basis ψ. The new basis is orthonormal and C is a unitary matrix.

   Minimizing with respect to $C^*_{p\alpha}$, remembering that $C^*_{p\alpha}$ and $C_{p\alpha}$ (and that the indices contain all single-particle quantum numbers including spin) are independent and defining

$$h^{HF}_{\alpha\gamma} = \langle\alpha|h|\gamma\rangle + \sum_{p}\sum_{\beta\delta} C^*_{p\beta}C_{p\delta}\langle\alpha\beta|V|\gamma\delta\rangle_{AS},$$

   show that you can write the Hartree-Fock equations as

$$\sum_{\gamma} h^{HF}_{\alpha\gamma}C_{p\gamma} = \epsilon^{\rm HF}_p C_{p\alpha}.$$

   Explain the meaning of the different terms and define the Hartree-Fock operator in second quantization. Write down its diagrammatic representation as well. The greek letters refer to the wave functions in the original basis while roman letters refer to the new basis.

   We will now set up the Hartree-Fock equations by varying the coefficients of the single-particle functions. The single-particle basis functions are defined as

$$\psi_p = \sum_{\lambda} C_{p\lambda}\psi_{\lambda}.$$

   where in our case $p = 1, 2, 3, 4$ and $\lambda = 1, 2, 3, 4$, that is the first four lowest single-particle orbits of Fig. 1. Set up the Hartree-Fock equations for this system by varying the coefficients $C_{p\lambda}$ and solve them for values of $g \in [-1, 1]$. Comment your results and compare with the exact solution. Compute the total ground state energy using a Hartree-Fock basis and comment your results. Does the Hartree-Fock single-particle basis differ from the original basis ϕ_λ for this interaction? Keep the Hartree-Fock ground-state energy and the Hartree-Fock Slater determinant for the comparison in Part III.

Part III: From Slater determinants to matrix product states and the DMRG (45 points)

In Part I the exact ground state was obtained as a vector in the space of Slater determinants with no broken pairs, and in Part II it was approximated by a single Slater determinant. The density matrix renormalization group offers a third parametrization of the same state, in which the amount of correlation that is kept is controlled by a single number, the bond dimension χ. In this part we rewrite the pairing model in this language and connect it to the two previous parts. The relevant background is the chapter on the density matrix renormalization group in the lecture notes; the programs `dmrg.py` in `doc/BookManybody/BookPrograms/chapterdmrg/` and the notebook `doc/LectureNotes/dmrg.ipynb` may be used freely, but everything asked for below can also be done with `numpy` alone.

Notation. We regard each level $p = 1, \ldots, L$ as a *site* of a chain, with the levels ordered by their energy. With no broken pairs a level is either empty or occupied by a pair, so each site has a local dimension $d = 2$ with basis states $|0\rangle$ and $|1\rangle$. The pair creation and annihilation operators of exercise 1 act on one site only, $\hat{P}_p^+|0\rangle_p = |1\rangle_p$ and $\hat{P}_p^-|1\rangle_p = |0\rangle_p$, and the number operator of the level is $\hat{n}_p = \hat{n}_{p+} + \hat{n}_{p-}$. The Slater determinants of exercise 2 are then product states $|n_1 n_2 \cdots n_L\rangle$ with $n_p \in \{0, 1\}$; for four levels and two pairs they are

$$|1100\rangle, \quad |1010\rangle, \quad |1001\rangle, \quad |0110\rangle, \quad |0101\rangle, \quad |0011\rangle,$$

where for example $|1100\rangle = \hat{P}_1^+ \hat{P}_2^+ |0\rangle$ is the reference state of exercise 4. If broken pairs are allowed, each site has instead $d = 4$ states, $|0\rangle$, $|+\rangle$, $|-\rangle$ and $|+-\rangle$.

Any state of the chain can be written as a *matrix product state* (MPS)

$$|\Psi\rangle = \sum_{n_1 \cdots n_L} A^{[1]n_1} A^{[2]n_2} \cdots A^{[L]n_L} \, |n_1 n_2 \cdots n_L\rangle,$$

where $A^{[p]n_p}$ is a $\chi_{p-1} \times \chi_p$ matrix for every value of n_p, with $\chi_0 = \chi_L = 1$ so that the product is a number. The χ_ℓ are the *bond dimensions*; the bond ℓ is the cut between site ℓ and site $\ell + 1$. The smallest possible χ_ℓ is the rank of the *coefficient matrix* of the cut,

$$C_{(n_1 \cdots n_\ell),(n_{\ell+1} \cdots n_L)} = \langle n_1 \cdots n_L | \Psi \rangle,$$

whose singular value decomposition $C = U \Sigma V^\dagger$ is the Schmidt decomposition $|\Psi\rangle = \sum_{k=1}^{\chi_\ell} \sigma_k |\ell_k\rangle |r_k\rangle$ across that cut. The *entanglement entropy* of the cut is $S_\ell = -\sum_k \sigma_k^2 \ln \sigma_k^2$ for a normalized state, and an MPS is in *left-canonical form* when $\sum_n (A^{[p]n})^\dagger A^{[p]n} = \mathbf{1}$ for every site p, which is what one obtains by performing the singular value decompositions from left to right and absorbing $\Sigma V^\dagger$ into the next site. A product state, such as a single Slater determinant in the level basis, has $\chi_\ell = 1$ for every cut and zero entanglement entropy.

In the same way an operator can be written as a *matrix product operator* (MPO)

$$\hat{H} = \sum_{\{n\},\{n'\}} W^{[1]n_1 n_1'} W^{[2]n_2 n_2'} \cdots W^{[L]n_L n_L'} \, |n_1 \cdots n_L\rangle\langle n_1' \cdots n_L'|,$$

where it is convenient to think of $W^{[p]}$ as a $D \times D$ matrix whose entries are operators acting on site p; the first site contributes only the first row and the last site only the last column, so that the product $W^{[1]} W^{[2]} \cdots W^{[L]}$ is a single operator on the chain. The number D is the bond dimension of the MPO. It is helpful to read the product as an automaton walking along the chain, whose state keeps track of which operators have been placed and are still waiting for their partner.

The DMRG is the variational minimization of $\langle \Psi | \hat{H} | \Psi \rangle / \langle \Psi | \Psi \rangle$ over all MPS with bond dimensions at most χ, carried out one or two sites at a time. With $\chi = 1$ it searches among product states over the levels; with χ large enough it reproduces the exact FCI state.

5. (10/100 points) **The Hamiltonian as a matrix product operator.** Starting from the pair form of the Hamiltonian derived in exercise 1,

$$\hat{H} = \sum_{p=1}^{L} \varepsilon_p \hat{n}_p - \frac{g}{2} \sum_{pq} \hat{P}_p^+ \hat{P}_q^-, \qquad \varepsilon_p = \xi(p-1),$$

describe the effect of each term when it acts on a state $|n_1 n_2 \cdots n_L\rangle$. Restrict to zero seniority, that is $d = 2$, and write down the 2×2 matrices representing $\hat{P}_p^+$, $\hat{P}_p^-$ and $\hat{n}_p$ in the basis $\{|0\rangle, |1\rangle\}$ of one site. Explain why the

MPO bond dimension is $D = 4$, that is, identify the four states of the automaton that builds the Hamiltonian, and explain why the bond dimension does not grow with the number of levels L even though the interaction couples every level to every other level. Write out the MPO matrix $W^{[p]}$ for a general level p and multiply out the full operator for the case of four levels, $W^{[1]}W^{[2]}W^{[3]}W^{[4]}$, where $W^{[1]}$ is the first row and $W^{[4]}$ the last column of the general matrix. Check that every term of $\hat{H}$ appears exactly once, and that the resulting operator, restricted to the states with two pairs, reproduces the 6×6 Hamiltonian matrix of exercise 2. Finally, what would the MPO be if each site has $d = 4$ states $|0\rangle$, $|+\rangle$, $|-\rangle$ and $|+-\rangle$?

6. (5/100 points) **Fermions.** The particles are fermions. How does this affect the MPO? Recall that if the $2L$ spin-orbitals are ordered along the chain, a fermion operator acting on orbital m carries the parity string $\prod_{m'<m}(-1)^{\hat{n}_{m'}}$ of all orbitals before it (Jordan-Wigner transformation). Discuss both the $d = 2$ and the $d = 4$ representation, and explain why the pairing Hamiltonian is a particularly simple case. Which additional terms in a Hamiltonian, for instance a one-body term $a^\dagger_{p\sigma}a_{q\sigma}$ with $p \neq q$ which moves a single particle between levels, would require the parity strings to appear explicitly in the MPO, and how would they appear?

7. (15/100 points) **Matrix product states by hand: the degenerate limit.** Set $\xi = 0$, so that all levels are degenerate and the Hamiltonian consists of the pairing interaction alone.

   (i) For four levels and two pairs, explain why the ground state is the equal-amplitude superposition of the six ways of placing two pairs,

   $$|\psi\rangle = \frac{1}{\sqrt{6}}\Big(|1100\rangle + |1010\rangle + |1001\rangle + |0110\rangle + |0101\rangle + |0011\rangle\Big),$$

   where $|0\rangle$ refers to an empty level and $|1\rangle$ to a paired level. What is its energy, and what are the other eigenvalues of the 6×6 matrix in this limit? Compare with the g-dependence found in exercise 2 for $\xi = 1$.

   (ii) Write down the coefficient matrix C for each of the three cuts of the chain, and construct the MPS in left-canonical form.

   (iii) What is the entanglement entropy of the middle cut, and how does it compare with the largest value that cut could carry? What is the entanglement entropy of the Hartree-Fock state of exercise 4?

   (iv) The middle bond carries $\chi = 3$ states. Say in words what distinguishes them, and use that to write down an MPS of 3×3 matrices (except at the edges) that generalizes to a model of L levels and two pairs. Close to the edges some of the three bond states are never used. Which, and why?

   (v) What are the bond dimensions χ_ℓ for each bond $\ell = 1, \ldots, L-1$ for N pairs in L degenerate levels? How do they compare with the number of Slater determinants $\binom{L}{N}$ that the FCI calculation of exercise 2 has to handle, and with the maximal bond dimension $\min(2^\ell, 2^{L-\ell})$ of an arbitrary state of the chain?

8. (15/100 points) **From FCI via Hartree-Fock to the DMRG.** Return to $\xi = 1$ and to the exact ground state of exercise 2 for $g \in [-1, 1]$, written as a vector in the basis of the six states $|n_1 n_2 n_3 n_4\rangle$ above.

   (i) For each of the three cuts, compute numerically the singular value decomposition of the coefficient matrix, the Schmidt weights σ_k^2 and the entanglement entropy S_ℓ. Plot the entropy of the middle cut as a function of g and compare with the value found in exercise 7 for $\xi = 0$. Comment on the limit $g \to 0$ and on the difference between $g > 0$ and $g < 0$. Which cut carries the most entanglement, and why?

   (ii) The Hartree-Fock state of exercise 4 is a single Slater determinant. Check whether it is a product state in the basis $|n_1 n_2 n_3 n_4\rangle$, and if so give its bond dimensions. Truncate the Schmidt decomposition of the exact state across the middle cut to $\chi = 1$ and to $\chi = 2$ Schmidt states, renormalize, and compute the energy expectation value $\langle\hat{H}\rangle$ and the discarded weight $\sum_{k>\chi}\sigma_k^2$ of the truncated states as functions of g. Compare with the exact energy of exercise 2, the truncated-CI energy of exercise 3 and the Hartree-Fock energy of exercise 4. Explain in which sense the $\chi = 1$ state and the Hartree-Fock state are both "best product states", and why they need not be the same state nor give the same energy.

   (iii) Build the MPO of exercise 5 numerically as an array of shape (D, D, d, d) and contract $W^{[1]}W^{[2]}W^{[3]}W^{[4]}$ to a 16×16 matrix. Verify that, restricted to the two-pair states, it reproduces the Hamiltonian matrix of exercise 2 for a few values of g. Then compute $\langle\Psi|\hat{H}|\Psi\rangle$ for the truncated MPS of the previous point by contracting the MPO with the MPS, and check that you obtain the same energies as above.

   If time permits, run a two-site DMRG (your own, or `dmrg.py`) for the four-level system with $\chi = 1, 2, 3$ and compare with the three previous points, and thereafter for a larger system, for example $L = 8$ levels and four

pairs or $L = 12$ levels and six pairs, where the seniority-zero FCI matrix is still small enough to diagonalize. Study the entanglement entropy across every bond as a function of g and comment on where along the chain the entanglement is concentrated. This last part is not required for full score.